

BO1 :

Identify the few factors that truly influence the result, even when most available information is irrelevant.

DSO1:

Detect and prioritize the small subset of informative variables in datasets dominated by noise or irrelevant features.

BO2:

Understand and use the combined influence of several factors that only make sense when considered together.

DSO2:

Capture complex multivariate interactions that do not exhibit strong marginal effects but drive the underlying data structure.

BO3 :

Increase the reliability of decisions when dealing with situations that involve many pieces of information but very few examples to learn from.

DSO3:

Improve predictive performance in high-dimensional settings where the number of features greatly exceeds the number of observations ($p \gg n$).

DSO1 — Prioritizing Informative Variables

Category	Details
Models to Use	- RLT with variable muting enabled - Optional checks: Random Forest or Lasso to confirm variable importance
Variables Involved	- Full feature set P at the start - Muted set (Pd): irrelevant variables removed over rounds - Protected set (P0): once selected, a variable is never removed - Final strong variables = highest VI scores
Key Parameters	- Muting rate: ~ 50% of remaining variables - Protected minimum (P0): 0–5 variables - Number of trees: ≈100 - Minimum node size: 2 - Embedded model: 100 trees, bootstrap ≈ 0.85

DSO2 — Capturing Multivariate Interactions

Category	Details
Models to Use	-RLT with linear combination splits activated - Reinforcement-based variable selection on - Optional baselines: Gradient Boosting or BART
Variables Involved	-Top VI variables used to create interaction features - Includes variables with no marginal effect, but strong joint effect - Protected set P0 keeps key interaction variables - Noise variables removed into Pd
Key Parameters	- Max variables in linear combo (k): 2–5 - VI threshold (α): ~0.25 - Random split quantile (q): 0.1–0.5 - Trees (M): ~100 Minimum node size (nmin): 2–5

DSO3 — Improving Performance in $p \gg n$ Settings

Category	Details
Models to Use	- RLT with aggressive muting - Optional comparison: ExtraTrees for speed
Variables Involved	- Full high-dimensional feature space P - Small strong subset p_s detected early - Muted set P_d grows quickly to remove noise - Protected set P₀ stabilizes selection when n is small
Key Parameters	- Muting rate: 70–80% of remaining variables - Protected minimum (P₀): usually not required - Minimum node size: 2 - Embedded model size: small/fast - Bootstrap ratio: ≈0.85 - Trees: ≈100